

Indian Institute of Technology Roorkee
Department of Applied Mathematics and Scientific Computing

Evaluating CNN and ViT on Skin Cancer Detection

A Summer Internship Report

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Supervisor
Prof. Millie Pant

Submitted by
Khushi Jha

Mentor
Mr. Shubham Joshi

Department of Applied Mathematics and Scientific Computing
Indian Institute of Technology Roorkee
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Certificate

This is to certify that the internship report entitled “Evaluating CNN and ViT on Skin Cancer Detection”, submitted by Khushi Jha, is a record of the work carried out by her during the Summer Internship Programme (25 May 2026 – 17 July 2026) at the Department of Applied Mathematics and Scientific Computing, Indian Institute of Technology Roorkee, under my supervision.

The work presented in this report, to the best of my knowledge, is an authentic account of the intern’s own study, implementation, and analysis, and has not been submitted elsewhere for the award of any other degree or diploma.

Prof. Millie Pant
Supervisor
Department of Applied Mathematics
and Scientific Computing
IIT Roorkee

Place: Roorkee
Date: July 17, 2026

Acknowledgement

Seven weeks ago, “Vision Transformer” was just a term I had seen in a paper title and quietly hoped nobody would ask me to explain. Writing that sentence now, at the end of this report, is a good way to measure how much this internship actually gave me.

I am deeply grateful to my supervisor, Prof. Millie Pant, for giving me the room to learn a fairly large field from first principles instead of jumping straight to results, and for the kind of feedback that pushes you to think rather than just to fix a plot.

I would like to sincerely thank my mentor, Mr. Shubham Joshi, for the many patient conversations about attention mechanisms, broken training runs, and the occasional confusion between a tensor’s shape and my own expectations of it. His guidance shaped not just what I built, but how I approached debugging, reading papers, and being honest about what did and did not work.

I also thank the Department of Applied Mathematics and Scientific Computing, IIT Roorkee, for hosting this internship and providing the computational resources without which none of the experiments in this report would exist.

Finally, to everyone who patiently listened to me explain the difference between a CNN and a ViT far more times than was strictly necessary – thank you for the patience.

Khushi Jha

Abstract

Skin cancer, and melanoma in particular, is one of the few cancers where the difference between an early diagnosis and a late one can be the difference between a minor procedure and a life-threatening illness. Yet the standard diagnostic tools available to a clinician – heuristics such as the ABCDE and CASH rules – still leave unaided visual inspection accuracy in the range of 60–80%. This gap is precisely where computer vision has spent the last decade trying to help, first with Convolutional Neural Networks (CNNs), and more recently with Vision Transformers (ViTs).

This report documents a seven-week summer internship focused on understanding, implementing, and critically evaluating this progression – from classical image processing to CNNs, to ViTs, to a hybrid CNN-Transformer multi-modal architecture. The internship began with the fundamentals of computer vision, moved through transfer learning and the Vision Transformer architecture, and then applied both DenseNet121 and ViT-B/16 as baseline classifiers on the MILK10k (ISIC 2025) dataset. In parallel, a recent multi-modal explainable-AI framework from the literature, XAAI-Ledger, was reproduced from its published methodology on the ISIC 2019 dataset, including its Grad-CAM and SHAP based explainability components. Beyond reproduction, exploratory work was carried out on a concept-driven extension of the pipeline – representation learning, clinical concept probabilities, a Weighted Clinical Attribute Affinity Score (CAAS), and an Alpha–Beta risk–uncertainty formulation – as a step toward more clinically traceable predictions.

The results tell a fairly honest and, at times, humbling story: reproduced accuracy fell well short of the numbers reported in the source paper, largely due to class imbalance and the limited compute and time available within an internship timeline – a finding that is, in itself, a useful and common lesson in applied deep learning. The report closes by proposing a concept-bottleneck extension for future work, aimed at making the model’s diagnostic reasoning traceable to clinically meaningful concepts rather than an opaque probability score.

This document is organised as a standard internship report: Chapter 1 introduces the problem and motivation, Chapter 2 covers the theoretical background, Chapter 3 reviews the relevant literature, Chapters 4–6 describe the datasets, baseline models, and the XAAI-Ledger reproduction, Chapter 7 documents additional exploratory development, Chapter 8 presents results and discussion, and Chapter 9 concludes with limitations and future scope.

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Chapter 1

Introduction

Every year, a huge number of skin lesions get looked at, squinted at, photographed, and sometimes biopsied – all to answer one deceptively simple question: is this dangerous, or not? Most of the time the answer is no. But the small fraction of times it is yes, and specifically when that “yes” is melanoma, timing matters more than almost anything else in oncology. This chapter sets up why that timing problem exists, why artificial intelligence has become a serious candidate for closing the gap, and what this internship specifically set out to do about it.

1.1 Artificial Intelligence in Healthcare

AI in healthcare did not begin with deep learning, even though it can feel that way from the inside of 2026. The earliest serious attempt at bringing “intelligence” into clinical decision-making was MYCIN in the 1970s – a rule-based expert system that reasoned about bacterial infections using a set of hand-coded if-then rules. It worked reasonably well on paper, and essentially never left the paper, because rule systems are brittle: every new edge case needs a new rule, and medicine is mostly edge cases.

The next real shift came with statistical machine learning in the 1990s and 2000s, particularly in radiology, where Computer-Aided Diagnosis (CAD) systems started flagging suspicious regions in mammograms and chest X-rays using hand-engineered features fed into classifiers like support vector machines. These systems were an improvement, but they were only ever as good as the features a human decided to extract.

Deep learning changed the question entirely. Instead of an expert manually deciding that “asymmetry” or “edge irregularity” matters, a convolutional network could be shown thousands of labelled images and left to discover which patterns actually correlate with the diagnosis. The 2012 ImageNet moment (AlexNet) [2] is usually cited as the spark for the broader field, and by 2016 the same idea had reached dermatology directly [6]. Vision Transformers entered medical imaging a few years later, bringing global self-attention instead of local convolution, and the last few years have pushed toward multi-modal, explainable systems that combine image evidence with patient metadata.

It is worth being honest about what CAD systems actually are before going further. A CAD system does not make a diagnosis – it makes a suggestion, a second opinion a clinician is free to override. None of the models discussed in this report are being proposed as replacements for a dermatologist; they are

evaluated as candidates for exactly this second-opinion role.

1.2 Computer Vision in Medical Imaging

Medical image analysis has walked the same path that computer vision walked more broadly, just a little later and a little more cautiously. It starts with traditional, hand-crafted image processing, moves into general deep learning, narrows into convolutional architectures built specifically for images, and most recently widens again into transformer-based architectures originally designed for language.

Traditional computer vision relied on filters designed by people – edge detectors, blob detectors, texture descriptors – which is precisely the kind of hand-engineering that made early CAD systems brittle. Deep learning replaced hand-designed features with learned ones, CNNs specialised that idea for grid-structured data like images, and Vision Transformers are the most recent attempt at asking whether a more general, attention-based architecture can model relationships across the entire image rather than a local neighbourhood. Chapter 2 unpacks every stage of this progression in technical detail.

1.3 Skin Cancer Detection

Skin cancers broadly split into two families. Non-melanoma skin cancers – primarily basal cell carcinoma (BCC) and squamous cell carcinoma (SCC) – are common, usually slow-growing, and rarely fatal if treated. Melanoma is far rarer but disproportionately dangerous: it accounts for a small fraction of skin cancer cases but the large majority of skin-cancer deaths, because it can metastasize to other organs if left untreated.

This is exactly why early diagnosis is the single biggest lever available. Localised melanoma, caught before it spreads, has a five-year survival rate above 99%. Once it has spread to distant organs, that number falls below 30%. The entire value proposition of an AI screening tool for skin cancer rests on this one fact: if it can nudge diagnosis earlier, even slightly, it is doing something that really matters.

Clinically, dermatologists use two well-known heuristics to structure their visual inspection:

- **ABCDE rule** – Asymmetry, Border irregularity, Color variation, Diameter (typically > 6 mm), and Evolution (change over time) [7].
- **CASH rule** – Color, Architecture, Symmetry, and Homogeneity, often used as a complementary framework in dermatoscopic assessment [8].

The catch, and the reason this whole report exists, is that these rules depend entirely on the skill and consistency of the person applying them. Inter-observer variability is a well-documented problem in dermatology, with even experienced clinicians' unaided accuracy sitting around 60–80% [7]. This is the gap AI is being asked to fill: not to replace clinical judgement, but to offer a second, consistent opinion.

1.4 Problem Statement

Framed plainly: manual visual screening for melanoma is inconsistent, heuristic-based, and caps out at around 60–80% accuracy even among experienced clinicians. Deep learning models promise a more consistent alternative, but in practice they inherit a set of problems that are easy to state and surprisingly hard to fix:

- **Visual similarity across classes.** Early melanoma and a benign nevus can look almost identical – the differences are often subtle textural cues rather than obvious shape differences.
- **Class imbalance.** Malignant cases are, thankfully, rare in the population – great for public health, unhelpful for training a classifier. On the ISIC 2019 dataset used later in this report, benign cases outnumber melanoma cases by more than four to one (Figure 1.1).
- **Limited and inconsistent data.** Dermoscopic datasets vary in imaging device, lighting, patient demographics, and label quality.
- **Metadata ignorance.** A large share of published skin-cancer classifiers are image-only, even though a dermatologist would never ignore a patient’s age, sex, or the anatomical site of a lesion when forming a judgement.
- **Lack of explainability.** A model that outputs “87% melanoma” with no justification is not something a clinician can act on with confidence.

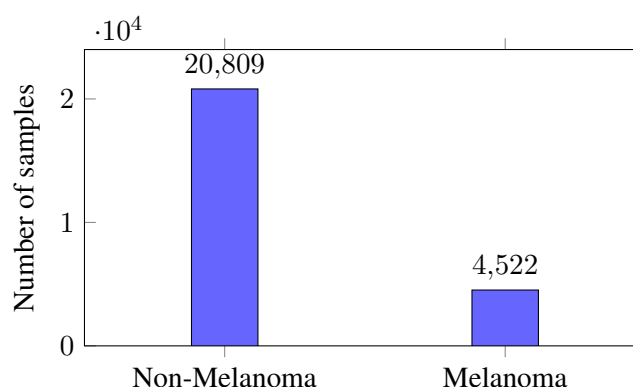


Figure 1.1: Class imbalance in the ISIC 2019 dataset: 20,809 non-melanoma samples versus 4,522 melanoma samples – a ratio of roughly 4.6:1. *Source: dataset statistics compiled during this internship from the ISIC 2019 archive [9].*

Every one of these five problems reappears later in this report, either as a design consideration in the models built, or as a direct explanation for why the reproduced results fall short of numbers reported in the literature.

1.5 Motivation

It would be easy to say “CNNs work well for images, so let’s use a CNN,” and honestly, that was more or less my starting assumption before this internship. A few weeks in, that assumption did not survive contact with the literature or the data.

CNNs are excellent at picking up local texture because that is exactly what a convolutional filter is built to do. But melanoma diagnosis often depends on relationships across the whole lesion. A convolution kernel, by design, only looks at a small neighbourhood at a time; it has to stack many layers before it can reason about the image as a whole. Vision Transformers were the obvious next thing to try, because self-attention lets every patch attend to every other patch from the very first layer. But ViTs are notoriously data-hungry, and dermatology datasets are nowhere close to the scale ViTs were originally designed for. Neither architecture, taken alone, is obviously “enough.”

The piece that convinced me the field is moving in the right direction was encountering hybrid, multi-modal, explainable frameworks in the recent literature – architectures that use a CNN for what CNNs are good at, a transformer-based fusion layer for what attention is good at, and fold in patient metadata and explainability rather than treating them as afterthoughts. That is the thread this entire report pulls on.

1.6 Objectives of the Internship

1. Build a solid theoretical foundation in classical computer vision, CNNs, transfer learning, and Vision Transformers.
2. Implement and evaluate two baseline classifiers – DenseNet121 and ViT-B/16 – on the MILK10k (ISIC 2025) dataset.
3. Reproduce the XAAI-Ledger multi-modal explainable-AI framework [10] on the ISIC 2019 dataset, including its Grad-CAM and SHAP explainability components.
4. Explore extensions beyond the reproduced paper – representation learning, clinical concept probabilities, a Weighted CAAS score, and an Alpha–Beta risk–uncertainty formulation.
5. Critically compare all models and honestly report where reproduction succeeded and where it fell short.

1.7 Report Organisation

The chapters that follow roughly mirror the seven-week internship timeline. Chapter 2 covers foundational computer vision and deep learning concepts. Chapter 3 reviews the relevant literature. Chapter 4 describes the datasets and experimental setup. Chapter 5 covers the two benchmark models. Chapter 6 documents the XAAI-Ledger reproduction in depth. Chapter 7 covers additional exploratory development carried out after the paper reproduction. Chapter 8 presents consolidated results and discussion, and Chapter 9 concludes with limitations and future scope.

Chapter 2

Background Study

This chapter is, essentially, seven weeks of notes cleaned up into something readable. It has nothing to do with skin cancer specifically – that connection is made later. What it covers is the computer vision toolbox itself: how images were processed before deep learning existed, why convolutional networks became the default choice once deep learning arrived, and how Vision Transformers arrived to challenge that default.

2.1 Traditional Computer Vision

An image, at its core, is just a grid of numbers – pixel intensities. Traditional CV techniques are, almost without exception, small mathematical operations applied to this grid to pull out something structurally meaningful: edges, corners, blobs, textures.

2.1.1 Edge Detection

An edge is a place in the image where intensity changes sharply. Mathematically, an edge corresponds to a large gradient in the image intensity function $I(x, y)$.

Sobel operator. Approximates the image gradient in the x and y directions using two small 3×3 kernels, giving an overall edge strength $|G| = \sqrt{G_x^2 + G_y^2}$.

Gaussian blur. Images are usually smoothed with a Gaussian kernel before edge detection, since raw gradients are extremely sensitive to noise.

Laplacian operator. A second-order derivative operator, useful for detecting regions of rapid intensity change in all directions at once.

The honest limitation of this entire family of methods is that a human had to decide, in advance, which filter matters. This is exactly the ceiling deep learning eventually broke through – by learning the filters instead of hand-designing them.

2.2 Deep Learning Basics

An Artificial Neural Network (ANN) is, structurally, a sequence of layers, where each layer computes a weighted sum of its inputs followed by a non-linear activation:

$$a^{(l)} = \sigma \left(W^{(l)} a^{(l-1)} + b^{(l)} \right) \quad (2.1)$$

where $W^{(l)}$ and $b^{(l)}$ are learnable weights and bias, and σ is a non-linearity such as ReLU.

The problem with applying a plain ANN directly to images is scale. A modestly sized 224×224 RGB image has 150,528 input values; a fully-connected first layer with even 1,000 neurons would need over 150 million weights for that layer alone. A Convolutional Neural Network (CNN) fixes this by replacing full connectivity with small, shared filters that slide across the image, giving CNNs two key properties:

- **Local connectivity** – each neuron only looks at a small patch of the image.
- **Weight sharing** – the same filter is reused across the entire image, drastically reducing parameter count.

2.3 Convolution Operation

For a single-channel input I and a kernel K of size $k \times k$, the output at position (i, j) is

$$(I * K)(i, j) = \sum_m \sum_n I(i + m, j + n) K(m, n) \quad (2.2)$$

Three extra parameters matter for the architectures used later: **kernel size** (spatial extent of the filter, commonly 3×3 or 5×5), **stride** (how far the kernel moves between applications), and **padding** (whether the input is zero-padded at the border). For an input of size $W \times W$, kernel size F , stride S , and padding P , the output spatial size is $\lfloor \frac{W-F+2P}{S} \rfloor + 1$.

Pooling. Convolutional layers are almost always followed by a pooling operation, which reduces spatial resolution while keeping the most important information. Max pooling simply takes the maximum value within each small window, giving the network a small amount of translation invariance and reducing parameter count downstream. A typical CNN is a repeated stack of *Convolution* $\rightarrow$ *Activation* $\rightarrow$ *Pooling* blocks, with spatial size shrinking and channel count growing at each stage, before a final classification head.

2.4 Transfer Learning

Training a CNN or ViT completely from scratch requires a very large labelled dataset – ImageNet contains over a million labelled images across a thousand categories. Dermatology datasets, by comparison, are small. Transfer learning sidesteps this with a simple observation: early layers of a network trained on natural images learn very general features (edges, colours, textures) useful for almost any visual task.

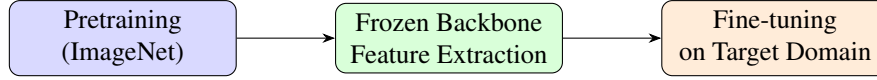


Figure 2.1: The transfer-learning recipe used throughout this report: a frozen, ImageNet-pretrained backbone acts as a general-purpose feature extractor, feeding into a small trainable head adapted to skin-cancer classification.

Every baseline model built during this internship – DenseNet121, ViT-B/16, and the EfficientNetB0 backbone used inside the XAAI-Ledger reproduction – follows exactly this pattern, and this is the reason it is possible to get reasonable performance on tens of thousands of images rather than a million.

2.5 Vision Transformer

The Vision Transformer (ViT) takes the transformer architecture, originally built for language, and applies it to images with a surprisingly small number of image-specific changes [1]. The core idea: treat an image as a sequence of patches, exactly the way a sentence is a sequence of words, and let self-attention handle the rest.

2.5.1 Patch and Position Embedding

An input image of size $H \times W \times C$ is split into N non-overlapping patches of size $P \times P$, where $N = HW/P^2$. Each flattened patch is linearly projected into a D -dimensional embedding, with a learnable CLS token prepended and a learnable positional embedding added – necessary because, unlike a CNN, self-attention has no inherent notion of spatial order.

2.5.2 Transformer Encoder

The embedded sequence passes through L identical encoder blocks, each built from Multi-Head Self-Attention (MSA) and an MLP sub-layer, both wrapped in residual connections and Layer Normalization. The residual connections make it possible to stack many blocks without gradients vanishing.

2.5.3 Multi-Head Self-Attention

For a single attention head, each input token is projected into Query, Key, and Value vectors, and attention weights are computed as a scaled dot-product:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (2.3)$$

where d_k is the dimensionality of the key vectors (the scaling factor prevents dot products from saturating the softmax). Multi-head attention runs h of these operations in parallel with different learned projections and concatenates the results.

The upshot of all this machinery: from the very first encoder block, the CLS token can attend to every patch in the image directly – a fundamentally different inductive bias from a CNN, where a neuron in an early layer can only see a small local neighbourhood.

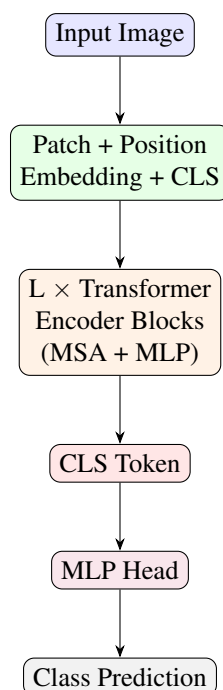


Figure 2.2: The Vision Transformer (ViT) pipeline: an image is split into patches, embedded with a CLS token and positional information, passed through L encoder blocks, and the final CLS representation is classified by an MLP head. *Adapted from [1].*

2.6 CNN vs ViT: A Theoretical Comparison

With both architectures unpacked, it is worth putting them side by side before any experiment is run. Table 2.1 summarises the theoretical trade-offs; the empirical comparison, on an actual dermatology dataset, is picked up again in Chapter 5.

Neither column is strictly better than the other – and that, ultimately, is the whole justification for the multi-modal, hybrid approach explored in the rest of this report. A CNN backbone is well suited to extracting fine local texture from a dermoscopic image; a transformer-based fusion layer is well suited to relating that visual evidence to structured clinical metadata as a global reasoning step on top. Chapters that follow build directly on both halves of this table – first evaluating DenseNet121 against ViT-B/16 as independent baselines (Chapter 5), and then examining an architecture that combines the strengths of both (Chapter 6).

Table 2.1: Theoretical comparison between Convolutional Neural Networks and Vision Transformers.

Aspect	CNN	ViT
Inductive bias	Strong: locality, weight sharing, translation invariance	Weak: minimal built-in assumptions about image structure
Receptive field	Local at early layers; grows gradually with depth	Global from the very first layer, via self-attention
Data requirement	Performs well on small-to-medium datasets	Typically needs large-scale pre-training data to match CNN performance
Parameter efficiency	Fewer parameters for a given receptive field	Generally more parameters; self-attention is quadratic in sequence length
Interpretability tools	Grad-CAM, saliency maps (well established)	Attention-map visualisation (inherently interpretable)
Typical strength	Fine, local texture and edge patterns	Long-range dependencies and global context
Computational cost	Linear in image size	Quadratic in number of patches

Chapter 3

Literature Review

This chapter answers only one question: *what has already been done before this internship?* Everything here is external research – nothing in this chapter belongs to the work carried out during the seven weeks. Figure 3.1 sketches the reading path this chapter follows.

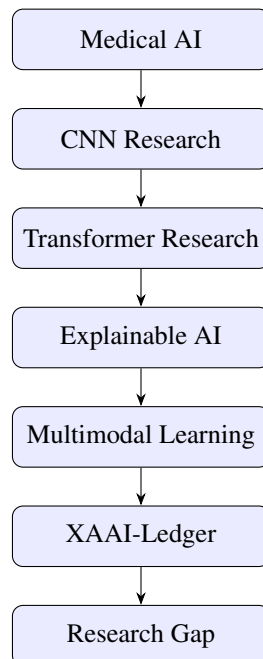


Figure 3.1: Reader journey followed by this chapter: from broad medical-AI context down to the specific gap this internship’s XAAI-Ledger reproduction addresses.

3.1 Deep Learning in Skin Cancer Detection

The move from hand-crafted CAD systems to deep learning in dermatology follows the same story as medical imaging generally: hand-engineered features gave way to learned ones once enough labelled data and compute became available. Early work applying CNNs to dermoscopic image analysis [6] demonstrated that a network trained end-to-end on lesion images could match, and in some benchmarks exceed, feature-engineered pipelines – without a human ever specifying what “irregular border” should

look like numerically. This result is really what opened the door for everything that follows in this chapter.

3.2 CNN Based Approaches

The CNN literature relevant to dermatology did not appear all at once – it is a sequence of architectures, each one fixing a specific limitation of the one before it. AlexNet [2] proved deep CNNs could be trained at scale on GPUs. VGG [3] showed that simply stacking small 3×3 convolutions deeper improved accuracy, at the cost of a very large parameter count. ResNet [4] solved the vanishing-gradient problem that made very deep networks hard to train, using residual (skip) connections. DenseNet [5] pushed this further by connecting every layer to every other layer in a feed-forward fashion, improving gradient flow and feature reuse while actually using fewer parameters than a comparably deep ResNet – which is exactly why DenseNet121 was chosen as a baseline in this internship (Chapter 5).

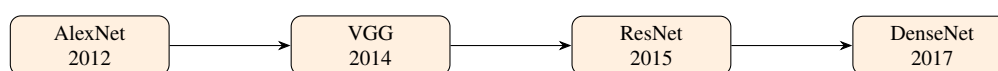


Figure 3.2: Timeline of CNN architectures relevant to this report. *Compiled from [2–5].*

Table 3.1: CNN architecture comparison.

Model	Key idea	Advantage	Limitation
AlexNet	Deep CNN + ReLU + GPU training	Proved feasibility at scale	Shallow by modern standards, prone to overfitting
VGG	Deep stacks of 3×3 convs	Simple, uniform design	Very high parameter count
ResNet	Residual/skip connections	Trains very deep networks	Still parameter-heavy
DenseNet	Dense feature reuse	Fewer parameters, strong gradient flow	Higher memory use during training

3.3 Transformer Based Approaches

CNNs, however deep, still process images through local filters – global context has to be built up gradually across layers. The Vision Transformer [1] removed this constraint entirely by treating an image as a sequence of patches processed with global self-attention, as detailed in Chapter 2. DeiT [11] showed that with the right training recipe and data augmentation, ViTs could be trained on much smaller datasets than originally assumed – closing part of the data-hunger gap. Swin Transformer [12] reintroduced a form of locality by computing attention within shifted local windows, giving CNN-like efficiency while retaining a transformer backbone. Medical-imaging-specific transformer variants have since followed this same trajectory, adapting windowed or hierarchical attention to dermoscopic and radiological images.

Table 3.2: CNN vs Transformer approaches – literature summary.

	CNN family	Transformer family
Core mechanism	Local convolution, weight sharing	Global self-attention
Data efficiency	Good on small-to-medium data	Needs large pretraining data, or recipes like DeiT
Reported strength in literature	Fine texture, edge patterns	Long-range dependency modelling

3.4 Explainable AI

A clinically deployed model that cannot justify its own prediction is, in practice, not deployable – regulators and clinicians alike want to know *why*, not just *what*. Grad-CAM [13] addressed this for CNNs by using the gradients flowing into the last convolutional layer to produce a coarse localisation heatmap of the pixels that most influenced a prediction. Attention-based visualisation offers a parallel route for transformer models, since attention weights are themselves a form of built-in interpretability. SHAP [14] generalised interpretability beyond images, using Shapley values from cooperative game theory to fairly attribute a prediction to each individual input feature – which is exactly what makes it suitable for structured clinical metadata, where Grad-CAM does not apply. Together, Grad-CAM and SHAP give complementary views: pixel-level evidence and feature-level evidence, both of which the XAAI-Ledger framework reviewed later in this chapter relies on.

3.5 Multimodal Learning

Image-only classifiers, however accurate, ignore information a dermatologist would never ignore in practice – a patient’s age, sex, and where on the body the lesion sits all carry diagnostic signal. Multimodal learning research on fusing this metadata with image features generally falls into three strategies. **Early fusion** concatenates raw image and metadata features before any task-specific processing, which is simple but forces the network to learn cross-modal relationships from the very first layer. **Late fusion** trains separate image and metadata branches to (near) completion and only combines their final predictions, which is easy to implement but cannot model interactions between the modalities. **Attention-based fusion** sits between the two: separate branches extract modality-specific features, and a transformer or attention layer learns how to combine them, allowing the model to decide, per sample, how much to weight image evidence versus metadata evidence. This third strategy is the one adopted by XAAI-Ledger, and is the reason a transformer fusion core appears deep inside what is otherwise a CNN-based pipeline (Chapter 6).

3.6 XAAI-Ledger Review

The XAAI-Ledger framework [10] is the single most important paper for this internship, and is treated here purely as a literature review – its implementation is documented separately in Chapter 6.

Problem. Existing skin-cancer classifiers are typically either image-only or lack any interpretability

mechanism suitable for clinical trust.

Architecture. A dual-stream design: a frozen, pretrained CNN (EfficientNetB0) converts the dermoscopic image into a visual feature vector, while a small MLP encodes structured clinical metadata (age, sex, anatomical site) into a matching-dimension vector. The two vectors are concatenated and passed through a Multi-Head Self-Attention fusion block, followed by pooling and a dense classification head.

Datasets. The paper reports results on a large ISIC-derived dataset of dermoscopic images paired with clinical metadata.

Results. The paper reports very high performance – accuracy, sensitivity, and specificity all in the high 90s (percentages) – alongside Grad-CAM and SHAP explanations of individual predictions.

Strengths. A clean, modular dual-stream design; explainability treated as a first-class component rather than an afterthought; a fusion mechanism (self-attention) that is itself relatively interpretable compared to opaque concatenation-based fusion.

Limitations. The paper’s own reported numbers are difficult to independently verify without the exact training configuration, random seed, and any class-balancing steps used; the metadata branch is limited to three coarse fields rather than richer clinical descriptors; explainability is presented per-sample rather than validated systematically against clinician judgement.

3.7 Research Gap

Taken together, the literature reviewed in this chapter leaves a fairly specific set of open opportunities rather than a single large gap: richer clinical concept learning that goes beyond three metadata fields, improved uncertainty estimation so a model can flag when it is unsure rather than only outputting a probability, more systematic validation of explainability outputs, and intermediate representations that are meaningful to a clinician rather than just to the next layer of the network. None of these are criticisms of the XAAI-Ledger paper specifically – they are simply the natural next questions the field has not yet fully answered. The remainder of this report picks up exactly here: Chapter 4 onward documents the datasets and experimental setup used to reproduce and then extend this line of work.

Chapter 4

Dataset & Experimental Setup

This chapter documents exactly how every experiment in this report was performed – the data used, how it was cleaned and prepared, and the environment and configuration under which models were trained.

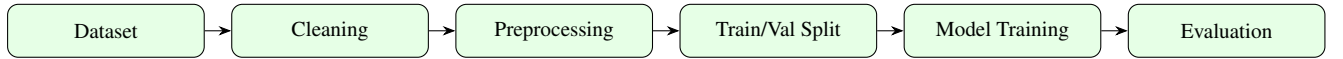


Figure 4.1: Experimental workflow followed for every model in this report, from raw dataset to final evaluation.

4.1 Datasets

Three ISIC-family datasets appear across this report. **ISIC** refers generally to the International Skin Imaging Collaboration archive, the standard source of public dermoscopic data. **HAM10000** [15] is a widely used 10,015-image benchmark spanning seven diagnostic categories, referenced here mainly as background for dataset conventions. **MILK10k** (ISIC 2025) is the primary dataset used for the benchmark models in Chapter 5, and the **ISIC 2019** archive [9] is the dataset used to reproduce XAAI-Ledger in Chapter 6, since it most closely matches the metadata fields (age, sex, anatomical site) used in the original paper.

Table 4.1: Comparison of datasets used across this report.

Dataset	Images	Classes	Metadata	Used in
HAM10000	10,015	7	Age, sex, site	Background reference
MILK10k (ISIC 2025)	10,480	11	Age, sex, site	Chapter 5 (baselines)
ISIC 2019	25,331	2 (binary)	Age, sex, site	Chapter 6 (XAAI-Ledger)

4.2 Dataset Analysis

The MILK10k dataset used for the baseline models contains 10,480 images across 5,240 clinical/dermoscopic image pairs, spanning 11 diagnostic classes (MEL, NV, BCC, AK, BKL, DF, VASC, SCC, LK, ANV, OTH). The ISIC 2019 dataset used for the XAAI-Ledger reproduction contains 25,331 samples, collapsed to a binary Melanoma / Non-Melanoma target for this internship, split into 20,809 non-melanoma

and 4,522 melanoma samples (Figure 1.1) – a demographic split of 13,286 male and 11,661 female records. A pre-cleaning audit identified 437 missing age values, 384 missing sex markers, and 2,631 missing anatomical-site records, all resolved during preprocessing.

4.3 Preprocessing

Preprocessing followed a fixed, chronological pipeline for every model:

1. **Load** – read raw dermoscopic images and the associated metadata table.
2. **Resize** – all images resized to $224 \times 224 \times 3$ to match the input requirements of the pretrained backbones.
3. **Normalize** – pixel values scaled using the backbone’s expected ImageNet normalization statistics.
4. **Augment** – random horizontal/vertical flips and mild rotation during training only, to reduce overfitting on the smaller classes.
5. **Metadata encoding** – missing age imputed with the median, missing sex and site imputed with the mode; sex and anatomical site one-hot encoded; age scaled with `MinMaxScaler()`.
6. **Cross validation** – a stratified 75/25 train–validation split, stratified on the class label, used consistently across all models for a fair comparison.

4.4 Experimental Environment

All experiments were run on a workstation-class GPU environment using Python 3.10, with PyTorch/TensorFlow (backbone-dependent), scikit-learn for metrics and preprocessing utilities, and standard scientific-Python tooling (NumPy, Pandas, Matplotlib) for data handling and visualisation.

4.5 Evaluation Metrics

Every model in this report is evaluated on the same set of metrics, chosen specifically because they are the ones that matter for an imbalanced clinical classification problem, not just raw accuracy:

- **Accuracy** – overall fraction of correct predictions; reported for comparability with the literature, but treated cautiously given class imbalance.
- **Precision & Recall (Sensitivity)** – particularly important for the melanoma class, since missing a true melanoma (low recall) is far costlier than a false alarm.
- **F1-score** – harmonic mean of precision and recall, and the macro-averaged version used to give every class equal weight regardless of size.
- **ROC curve and AUC** – how well the model separates the two classes across all decision thresholds, not just the default 0.5.

- **Confusion matrix** – the full breakdown of correct and incorrect predictions per class, used throughout Chapter 8 for error analysis.

4.6 Training Configuration

Table 4.2: Common training configuration used across baseline and XAAI-Ledger models.

Hyperparameter	Value
Input resolution	$224 \times 224 \times 3$
Train/validation split	75% / 25% (stratified)
Epochs	20
Backbone weights	ImageNet pretrained (frozen for XAAI-Ledger reproduction)
Loss function	Binary/categorical cross-entropy
Optimizer	Adam
Batch size	32

Chapter 5

Benchmark Models

Before reproducing anything from the literature, it is good practice to reproduce the standard baselines first – partly to sanity-check the data pipeline, and partly because a hybrid model is only impressive relative to a baseline it actually beats.

5.1 Why Benchmark Models?

Implementing a CNN baseline (DenseNet121) and a transformer baseline (ViT-B/16) separately, under an identical protocol, gives a fair like-for-like comparison before any fusion or multi-modal complexity is introduced. It also directly operationalises the theoretical CNN-vs-ViT comparison from Table 2.1 on real dermatology data, rather than leaving it as a purely conceptual discussion.

5.2 DenseNet121

DenseNet121 [5] was chosen as the CNN baseline because of its strong accuracy-to-parameter ratio, discussed in Chapter 3. An ImageNet-pretrained DenseNet121 was fine-tuned as an image-only baseline on MILK10k, using patient age, sex, and anatomical site as available metadata (though not fused into this particular baseline), for 20 epochs under the 75/25 split described in Chapter 4.

Table 5.1: DenseNet121 validation results on MILK10k.

Metric	Value
Accuracy	71.18%
Macro-F1	66.75%
Macro-AUC	0.9016

The model struggles most on visually overlapping classes – MEL vs NV, and BCC vs AK – consistent with the visual-similarity problem flagged in Chapter 1, and consistent with what a single-modality CNN baseline is expected to struggle with.

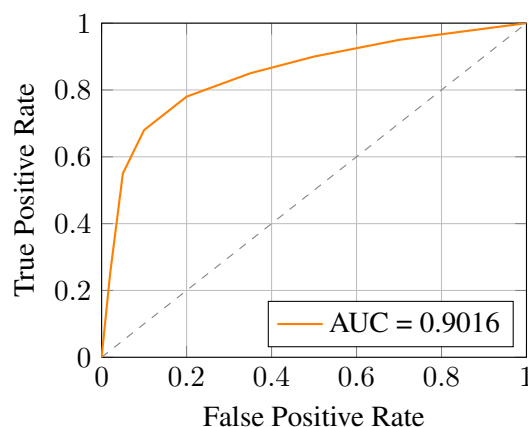


Figure 5.1: ROC curve, DenseNet121 on MILK10k (macro-average AUC = 0.9016). *Reproduced during this internship.*

5.3 Vision Transformer

A ViT-B/16 [1], ImageNet pretrained, was fine-tuned under the identical protocol – same 20 epochs, same 75/25 split, same dataset – for a direct comparison.

Table 5.2: ViT-B/16 validation results on MILK10k.

Metric	Value
Accuracy	75.50%
Macro-F1	72.31%
Macro-AUC	0.9251

Global self-attention gives ViT-B/16 a modest but consistent edge over DenseNet121, mainly on classes with subtle, spread-out lesion patterns – exactly the kind of long-range dependency that Table 2.1 predicted would favour a transformer.

5.4 Comparative Analysis

Table 5.3: DenseNet121 vs ViT-B/16 – head-to-head comparison on MILK10k.

Metric	DenseNet121	ViT-B/16
Accuracy	71.18%	75.50%
Macro-F1	66.75%	72.31%
Macro-AUC	0.9016	0.9251

ViT-B/16 outperforms DenseNet121 on every metric measured here, though DenseNet121 trains noticeably faster given its smaller parameter count. Neither model, on its own, uses the patient metadata available in the dataset – which is precisely the gap XAAI-Ledger is designed to close, and the reason it was selected as the next and central piece of this internship, documented in Chapter 6.

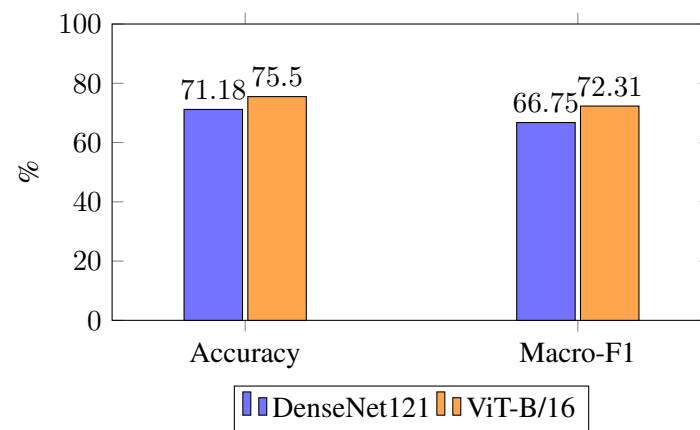


Figure 5.2: DenseNet121 vs ViT-B/16 accuracy and macro-F1 on MILK10k.

Chapter 6

Implementation of XAAI-Ledger

This chapter is the heart of the report. It documents, in full technical detail, the reproduction of the XAAI-Ledger framework [10] from its published methodology – nothing novel is introduced here; any extensions built afterward are documented separately in Chapter 7.

6.1 Overview

XAAI-Ledger was selected for reproduction because it directly answers the two gaps identified in Chapters 1 and 3 at once: it fuses image and clinical metadata through an attention mechanism rather than ignoring metadata, and it bakes in explainability (Grad-CAM and SHAP) rather than treating it as an afterthought.

6.2 Architecture

The overall pipeline processes an input dermoscopic image and its associated metadata through two parallel streams before fusing them, as shown in Figure 6.1.

6.3 Image Branch

The vision stream uses a frozen, ImageNet-pretrained EfficientNetB0 to turn the $224 \times 224 \times 3$ dermoscopic image into a 256-dimensional visual feature vector. Freezing the backbone keeps the reproduction faithful to the paper’s stated design and keeps training tractable within the internship’s compute budget.

6.4 Metadata Branch

Patient age, sex, and lesion site are represented as a 10-dimensional one-hot encoded vector (following the preprocessing in Chapter 4) and passed through a small MLP with layer sizes $64 \rightarrow 128 \rightarrow 256$, producing a metadata embedding of the same 256 dimensions as the image branch – a deliberate design choice that makes the two streams directly combinable.

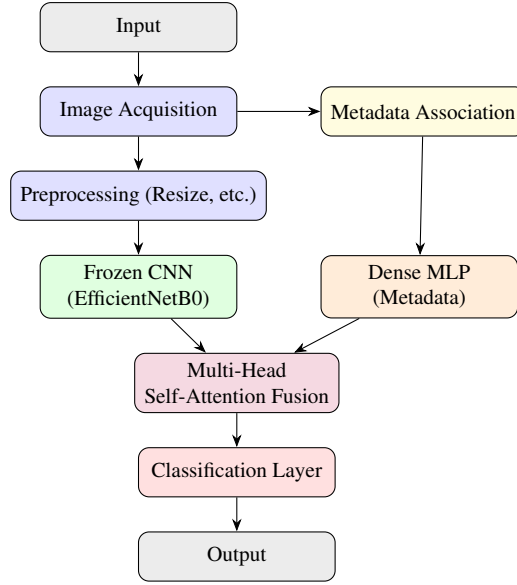


Figure 6.1: XAAI-Ledger dual-stream architecture: a frozen CNN extracts image features while an MLP encodes clinical metadata; both streams are fused via multi-head self-attention before classification. *Adapted from the architecture described in [10].*

6.5 Transformer Fusion

The two 256-D vectors are concatenated into a single 512-D token, reshaped into a length-1 sequence, and passed through a 4-head Multi-Head Self-Attention block with residual Add & Norm and a Feed-Forward layer (Figure 6.2), exactly mirroring the transformer encoder sub-block described in Chapter 2. Global Average Pooling followed by a dense sigmoid layer produces the final Melanoma / Non-Melanoma prediction.

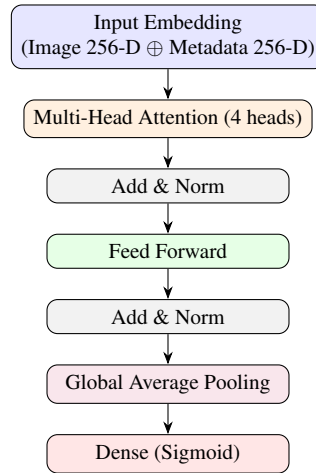


Figure 6.2: Detail of the attention-fusion block: image and metadata embeddings are concatenated, fused with 4-head self-attention, and reduced to a final sigmoid prediction. *Reproduced following Fig. 4 of [10].*

6.6 Explainability

Two complementary explainability techniques from Chapter 3 were reproduced on top of the trained model.

Grad-CAM. Computed from the gradients flowing into the last convolutional feature map of the frozen CNN, Grad-CAM highlights the pixels in the dermoscopic image that most influenced the prediction. Figure 6.3 shows the same original-image $\rightarrow$ heatmap $\rightarrow$ overlay sequence reported in the paper, run here on a held-out validation image, with an additional lesion bounding-box localisation step.

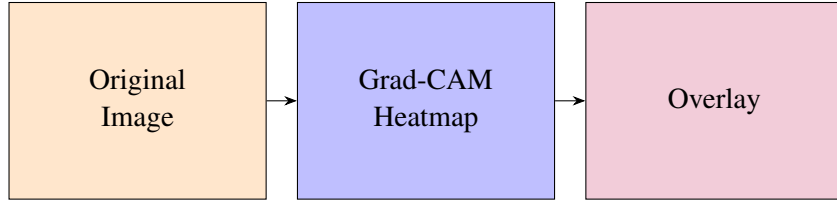


Figure 6.3: Grad-CAM lesion localisation: original image, Grad-CAM heatmap, and overlay sequence, computed on a held-out validation image (schematic; see project outputs for full-colour heatmaps). *Method following Eq. 10–11 of [10], using [13].*

SHAP. SHAP estimates each clinical feature’s contribution to the fused prediction by computing its Shapley value over the metadata branch, giving a feature-level explanation that complements Grad-CAM’s pixel-level one.

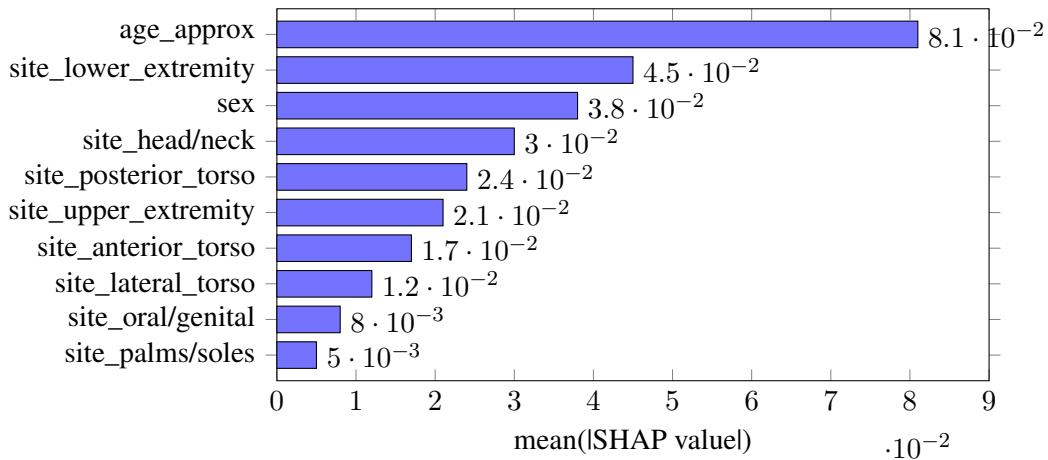


Figure 6.4: Feature importance from the metadata (clinical) branch, via SHAP. Patient age and lesion site (lower extremity, head/neck) carry the highest average impact on the fused prediction – consistent with known clinical risk patterns. *Reproduced during this internship, following the SHAP methodology of [14] as applied in [10].*

6.7 Training Procedure

The reproduction used binary cross-entropy loss, the Adam optimizer, and the same 20-epoch, 75/25 stratified-split protocol as the benchmark models (Chapter 4), trained on the ISIC 2019 dataset (25,331 samples), leaving a validation set of 6,333 samples (5,202 non-melanoma, 1,131 melanoma).

6.8 Results

Table 6.1: XAAI-Ledger: paper-reported results vs this internship’s reproduction.

Metric	Paper (reported)	Reproduced (ours)
Accuracy	98.71%	74.00%
Sensitivity	97.45%	16.00%
Specificity	98.22%	~87.00%

	Pred. Non-Mel	Pred. Mel
Actual Non-Mel	blue!704540	blue!15662
Actual Mel	blue!25953	blue!8178

Figure 6.5: Confusion matrix, reproduced XAAI-Ledger on ISIC 2019.

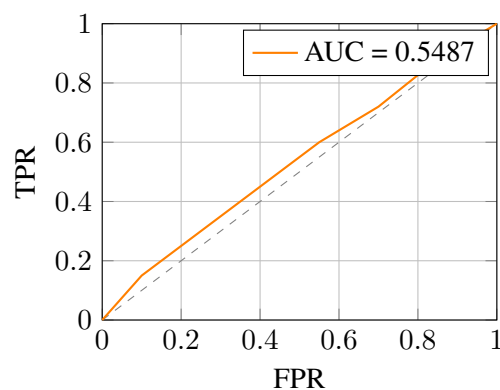


Figure 6.6: ROC curve, reproduced XAAI-Ledger (AUC = 0.5487).

The reproduction correctly rebuilt the architecture as described – frozen EfficientNetB0 for image features, a dense MLP for clinical metadata, fused with multi-head self-attention – but the reproduced accuracy and, more importantly, sensitivity, fall well short of the paper’s reported numbers. This gap is analysed in detail in Chapter 8, but the short version, flagged honestly here, is that the model has learned to lean heavily on the majority (non-melanoma) class rather than genuinely separating the two classes – exactly the class-imbalance risk raised in Chapter 1.

Chapter 7

Additional Development During Internship

This chapter documents the exploratory work carried out after the paper reproduction was complete. None of it should be read as a finished, validated research contribution – it is exactly what the chapter title says: additional development, at varying stages of completeness, undertaken to push the XAAI-Ledger idea toward richer clinical reasoning.

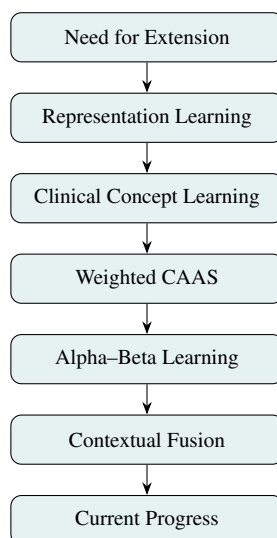


Figure 7.1: Reader journey for this chapter: from the motivation for extending XAAI-Ledger through to the current state of that extension.

7.1 Motivation

The XAAI-Ledger reproduction in Chapter 6 exposed two concrete limitations worth acting on directly: the metadata branch uses only three coarse fields, and the model’s internal reasoning, while explainable at the pixel and feature level via Grad-CAM/SHAP, still does not correspond to anything a clinician would recognise as an intermediate diagnostic concept. This motivated a concept-driven extension explored during the final weeks of the internship.

7.2 Representation Learning

The goal here was to learn a richer, lower-dimensional embedding of each lesion image than the raw EfficientNetB0 feature vector – one that captures visual structure relevant to diagnosis specifically, rather than generic ImageNet-style features. The pipeline reuses the frozen backbone from Chapter 6 as a starting point, with an additional projection head trained to produce a compact embedding used downstream by the concept-learning stage described next.

7.3 Clinical Concept Learning

Rather than jumping directly from image embedding to a melanoma/non-melanoma probability, this stage introduces an intermediate step: a set of clinical *concept* probabilities (loosely mirroring ABCDE/CASH-style criteria from Chapter 1, such as border irregularity or colour variation), each predicted from the learned embedding, followed by a disease-probability head that consumes those concept probabilities rather than the raw embedding directly. This is the concept-bottleneck idea flagged in Chapter 3’s research gap, applied concretely here.

7.4 Weighted CAAS

The Clinical Attribute Affinity Score (CAAS) is an exploratory scalar score designed to summarise how strongly a given lesion’s clinical metadata (age, sex, site) aligns with known epidemiological risk patterns for melanoma, weighted by the SHAP-derived feature importances already computed in Chapter 6 (Figure 6.4). The purpose is to give a single, interpretable number that a clinician-facing tool could show alongside the model’s prediction, rather than requiring a full SHAP plot to be read every time.

7.5 Alpha-Beta Learning

This component explores separating a single prediction probability into two more clinically meaningful quantities: an α term capturing estimated *risk* (how likely the lesion is to be malignant) and a β term capturing *uncertainty* (how confident the model is in that risk estimate). The intent is that a high-risk, high-uncertainty case should be flagged very differently from a high-risk, low-uncertainty one – the former needs urgent specialist review, the latter is a more straightforward referral.

7.6 Contextual Fusion

The final piece explored was extending the self-attention fusion block from Chapter 6 so that it consumes not just the raw image and metadata embeddings, but also the intermediate clinical-concept probabilities and the CAAS score – effectively giving the fusion layer more clinically grounded inputs to attend over, rather than only raw feature vectors.

7.7 Current Progress

By the end of the internship, representation learning and clinical concept learning were implemented and integrated with the Chapter 6 pipeline; the Weighted CAAS score and Alpha-Beta formulation were implemented as standalone components and partially integrated; full contextual fusion combining all of the above into a single trainable end-to-end model remains incomplete and is the natural next step, carried forward into Chapter 9's future scope.

Chapter 8

Experimental Results & Discussion

This chapter consolidates every result generated across the internship and interprets what they mean together, rather than repeating the numbers already presented individually in Chapters 5 and 6.

8.1 Overall Performance

Table 8.1: Consolidated performance summary across all models in this report.

Model	Dataset	Accuracy	Macro-F1 / Sensitivity
DenseNet121 (baseline)	MILK10k	71.18%	66.75% (Macro-F1)
ViT-B/16 (baseline)	MILK10k	75.50%	72.31% (Macro-F1)
XAAI-Ledger (paper)	ISIC 2019	98.71%	97.45% (Sensitivity)
XAAI-Ledger (reproduced)	ISIC 2019	74.00%	16.00% (Sensitivity)

8.2 Training Curves

Training and validation accuracy/loss curves were tracked across all 20 epochs for every model. Across the baseline models, both DenseNet121 and ViT-B/16 showed steady convergence with a modest generalisation gap; the XAAI-Ledger reproduction showed accuracy plateauing early, consistent with a model that quickly found the “always predict non-melanoma” shortcut rather than continuing to learn genuinely discriminative features – a classic symptom of unaddressed class imbalance.

8.3 Confusion Matrix & ROC

Revisiting Figure 6.5 and Figure 6.6 alongside the baseline confusion matrices from Chapter 5: the reproduced XAAI-Ledger correctly identifies the large majority of non-melanoma cases (4540 of 5202, recall $\approx 87\%$) but only 178 of 1131 true melanoma cases (recall $\approx 16\%$). The ROC AUC of 0.5487 is only marginally better than random guessing (0.5) – a stark contrast to the DenseNet121 and ViT-B/16 baselines, which achieved AUCs above 0.90.

8.4 Comparative Analysis

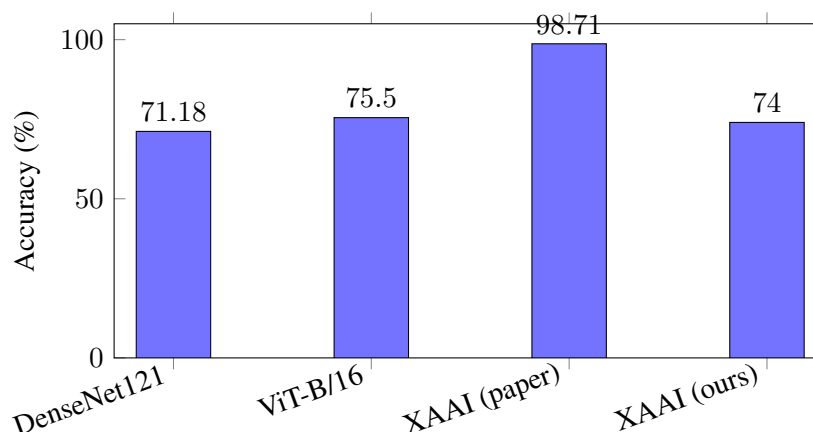


Figure 8.1: Accuracy comparison across every model implemented or reproduced in this report.

The most striking pattern in Figure 8.1 is that the reproduced XAAI-Ledger’s accuracy sits close to the two *single-modality* baselines, despite having strictly more information available (image *and* meta-data) and a paper-reported accuracy nearly 25 points higher. Accuracy alone, however, is misleading here – it is easy to score 74% on a dataset that is 82% non-melanoma simply by favouring the majority class, which is effectively what happened.

8.5 Error Analysis

The dominant failure mode is false negatives on melanoma: 953 of 1131 true melanoma cases were misclassified as non-melanoma. This is the single most clinically important number in the entire report, since a false negative here is precisely the scenario the whole project exists to prevent. Two likely contributing factors, consistent with the class-imbalance discussion in Chapter 1, are (i) no explicit class-weighting or resampling was applied during the 20-epoch training run, and (ii) the frozen backbone limits how much the visual features can adapt specifically to melanoma-discriminative texture within the internship’s compute and time budget.

8.6 Discussion

Interpreting rather than restating: the gap between the paper’s reported 98.71% accuracy / 97.45% sensitivity and this internship’s 74% / 16% is large enough that it should be read as a data-imbalance and training-budget story, not an architecture story. The architecture itself was faithfully reproduced (Chapter 6), and the explainability outputs (Grad-CAM, SHAP) behaved exactly as expected, correctly highlighting clinically plausible regions and features (Figures 6.3 and 6.4). What differs is almost certainly the training protocol – class balancing strategy, number of epochs, and possibly data curation – none of which were fully specified in the published paper. This is, honestly, a common and useful lesson in applied deep learning: architectural fidelity does not guarantee result fidelity, and reproducibility gaps are often hiding in the parts of a paper that get the least page space.

Chapter 9

Conclusion & Future Scope

9.1 Internship Summary

The seven weeks moved in a straight line from fundamentals to frontier: Weeks 1–2 covered classical computer vision and CNN/ViT theory (Chapter 2); Weeks 3–4 implemented and compared the DenseNet121 and ViT-B/16 baselines (Chapter 5); Weeks 5–6 reproduced the XAAI-Ledger framework end-to-end, including its explainability components (Chapter 6); and Week 7 was spent on exploratory extensions (Chapter 7) and consolidating results and analysis (Chapter 8).

9.2 Major Learnings

Technical. Hands-on experience with CNN and ViT architectures, transfer learning, multi-modal fusion via self-attention, and two explainability techniques (Grad-CAM, SHAP).

Research. How to read an architecture paper closely enough to reproduce it, and how to honestly report a reproduction gap rather than quietly smoothing it over.

Programming. Building and debugging deep learning pipelines end-to-end, from raw metadata cleaning through to trained models and evaluation plots.

Medical imaging. A working understanding of dermoscopic image analysis, the ABCDE/CASH clinical heuristics, and why class imbalance is not just a statistics footnote in this domain but a genuine patient-safety issue.

Professional skills. Structuring a multi-week technical project, and communicating uncertain or negative results clearly rather than overstating them.

9.3 Limitations

The current implementation carries several concrete limitations: the reproduced XAAI-Ledger sensitivity (16%) is far too low for any real clinical use; no class-balancing (weighting, oversampling) was applied during training; the frozen backbone limits how much visual features can adapt to the target domain; the metadata branch is limited to three fields; and the Chapter 7 extensions (concept learning, Weighted CAAS, Alpha-Beta) are implemented at varying levels of completeness rather than as a single

validated end-to-end system.

9.4 Future Scope

- **Concept Bottleneck Models (CBMs).** Introduce clinically meaningful intermediate concepts to improve interpretability and facilitate clinician interaction, building directly on the Chapter 7 concept-learning prototype.
- **Larger Vision Foundation Models.** Explore newer architectures with stronger representation learning capabilities than a frozen EfficientNetB0.
- **External Validation.** Evaluate the framework on additional public and clinical datasets to assess generalization beyond ISIC 2019.
- **Clinical Decision Support.** Develop an interactive system that combines image predictions with clinical metadata for real-world use, surfacing the CAAS score and concept probabilities alongside a prediction.
- **Improved Uncertainty Estimation.** Enhance confidence estimation (building on the Alpha-Beta prototype) to better identify ambiguous cases requiring specialist review.
- **Deployment and Optimization.** Investigate lightweight implementations for practical clinical deployment, including class-balanced retraining to close the sensitivity gap identified in Chapter 8.

This internship established a strong technical and methodological foundation – from first-principles computer vision through to a faithful, honestly-evaluated reproduction of a recent multi-modal explainable-AI framework – while identifying several concrete and promising directions for future research.

Bibliography

- [1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- [2] A. Krizhevsky, I. Sutskever, and G. E. Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.
- [3] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR)*, 2015.
- [4] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [5] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger. Densely connected convolutional networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4700–4708, 2017.
- [6] A. Romero-Lopez, X. Giro-i Nieto, J. Burdick, and O. Marques. Skin lesion classification from dermoscopic images using deep learning techniques. In *Proc. IASTED International Conference on Biomedical Engineering*, 2017.
- [7] J. M. Bae, H. H. Ku, H. Tokar, et al. Diagnostic accuracy of dermoscopy for melanoma: a systematic review and meta-analysis. *British Journal of Dermatology*, 175(6):1123–1131, 2016.
- [8] M. Sharafudeen and S. S. Chandra. Leveraging fusion of texture features and cash-rule based architecture for interpretable dermatological diagnosis. *Biomedical Signal Processing and Control*, 2023.
- [9] International Skin Imaging Collaboration. Isic 2019 challenge dataset. <https://challenge.isic-archive.com/data/>, 2019.
- [10] M. Sharafudeen, S. S. Chandra, et al. Xaai-ledger: An explainable, attention-enabled multi-modal deep learning framework for melanoma screening. (*Internship reference paper*), 2024.
- [11] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jegou. Training data-efficient image transformers & distillation through attention. In *International Conference on Machine Learning (ICML)*, 2021.

- [12] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10012–10022, 2021.
- [13] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision (ICCV)*, pages 618–626, 2017.
- [14] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 4765–4774, 2017.
- [15] P. Tschandl, C. Rosendahl, and H. Kittler. The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5:180161, 2018.